The Domain Modules

Instances of the Constitutional Runtime

The compilation discipline for the module layer: the module schema and its nine slots, the compilation contract with its acceptance tests, the module registry and composition rules, and five exemplar modules — each an instance of the constitutional runtime, not a separate invention universe.

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C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 11, the final paper of the eleven-paper research program defined by the SVT Architect Analysis file. Papers 1–7 built the machine and its laws; Papers 8–10 taught it to survive its infrastructure, price its resources, and govern its own creativity. The kernel is finished. What remains is the source's last instruction: only after the kernel exists should the domain modules P71–P90 be compiled against it, and each module becomes an instance of the constitutional runtime, not a separate invention universe. The source names five exemplar modules — P71 Pediatric Guardian, P72 Maternal Protector, P73 Palliative Navigator, P74 Emergency Architect, P75 AMR Fighter — each as a sum of domain facets, and closes with the sentence this paper exists to make true: *one constitutional mechanism, multiple empirical domains* VERIFIED-FROM-SOURCE.

Two interpretive decisions are recorded here and defended in the body. First, the source's facet sums (e.g., P73 = Objective Transition + Autonomy + Suffering + End-of-Life) are read as *compositions against predecessor papers,* not as feature lists: "Objective Transition" in P73's sum means Paper 7's guarded engine instantiated for the palliative objective space, with everything that paper's machinery already guarantees FORMALIZED-HEREIN. Second, because the modules are generated artifacts in the constitutional sense, the obligations Paper 10 recorded for this paper are discharged explicitly: every module enters the lifecycle at state G with zero authority, ships with its survivability dossier and constraint objects, and climbs the gauntlet like everything else the architecture has ever made FORMALIZED-HEREIN.

0.2 Notation

All notation of Papers 1–10 is retained. New here: M-OBJ — the module object with its nine slots; Δ_{dom} — the domain delta, the complete set of domain-specific content a module adds; Σ_{dom} — the module's domain safety properties, disjointly extending the kernel's Σ; $compat(m, K)$ — the compilation predicate of §3; the acceptance battery A-m — the nine tests of Table 3-1. Load-bearing inheritances: Paper 10's lifecycle automaton Σ_G governs every module; Paper 9's constraint structure governs every module's feasibility floor; Paper 8's lattice governs every module's degradation; Paper 7's guard governs every module's objectives.

0.3 Epistemic status convention

Every claim carries one of four tags, as fixed in Paper 1 and enforced throughout: VERIFIED-FROM-SOURCE for statements present in the source architecture; FORMALIZED-HEREIN for precise statements constructed in this document from source intent; CONJECTURE for hypotheses the architecture suggests but does not yet entail; UNKNOWN for open questions stated without a proposed answer. The falsification register in §12 lists the conditions under which each load-bearing claim would fail. Nothing in this paper is asserted beyond its tag.

0.4 Authority boundary

C0 informational. This document specifies the discipline by which domain modules are compiled against the kernel; it compiles none, certifies none, deploys none. The five module deltas of §§5–9 are specifications, not clinical artifacts: they name obligations, not practices. The paper tier fully specifies this paper: a module schema is a document, and a document treats no patient.

0.5 The thesis in one sentence

A module is not a product built on the kernel but a reading of the kernel into a domain — physiology, development, suffering, scarcity, ecology — and the constitution's last promise is that the reading is total: nothing in the module escapes the laws, and nothing in the laws is waived for the module.

TABLE OF CONTENTS

Part I — The Problem	
1	The Last Mile Is Constitutional, Not Technical
2	The Module Schema
Part II — The Invariant	
3	The Compilation Contract
4	The Module Registry and Composition
Part III — The Five Deltas	
5	P71 Pediatric Guardian
6	P72 Maternal Protector
7	P73 Palliative Navigator
8	P74 Emergency Architect
9	P75 AMR Fighter
Part IV — Evidence and Governance	
10	Falsifiable Hypotheses (H71–H77) and Verification
11	Reconciliation with Papers 1–10 and PRIME
12	Output Contract
13	References and Program Colophon
Appendices	
A	Module Compilation Checklist and Property Index
B	The Module Dossier Schema

PART I

The Problem

Why the module layer is where constitutions go to die, and the schema that keeps this one alive.

1 The Last Mile Is Constitutional, Not Technical

The kernel is finished. Ten papers of machinery stand between this sentence and the blank page the program began with: gates and transitions, composition and absence, audit and evidence, objectives, survival, feasibility, generation. Every one of those papers was written in the subjunctive of engineering — what the architecture *would* do, *would* refuse, *would* record. The module layer is where the subjunctive meets the indicative: a ward, a learner, a patient, a clock. It is also where every governance architecture in the field's history has quietly died — not in the kernel, where the abstractions are clean and the incentives aligned, but in the last mile, where a deployment team with a deadline discovers that the constitution is negotiable in exactly one direction FORMALIZED-HEREIN.

1.1 The source's ordering, and why it is law

The source's first line for this paper is an ordering constraint: *only after the kernel exists* should the modules be compiled against it VERIFIED-FROM-SOURCE. The ordering is not project management; it is the constitutional claim itself. A module built before its kernel is a module that will grow its own governance — the separate invention universe the source forbids — because every domain problem the kernel was built to solve will be solved again, worse, in the module's own idiom: authority by convention, evidence by folklore, absence by silence. The field's landscape is littered with these universes: clinical decision tools with their own override cultures, educational platforms with their own drift, each one a small constitution written in a hurry and amended in secret. The source's second line states the alternative as a definition: each module is an *instance of the constitutional runtime* VERIFIED-FROM-SOURCE. This paper's work is to give "instance" a semantics strong enough to refuse the first universe every time it is offered.

1.2 Instantiation, not integration

The word the industry would reach for is *integration*: the kernel is a platform, the modules are apps, and the platform's rules bind the apps as terms of service bind — loosely, revocably, and mostly at onboarding. This paper's word is *instantiation*, and the difference is the whole paper. Integration is a relationship between two artifacts; instantiation is a relationship between a law and a world. An integrated module may violate its platform's policies and remain integrated; an instantiated module that violates the constitution has a *type error* — it is not a bad module, it is not a module at all, and the compilation contract of §3 exists to make that distinction operational: compat(m, K) either holds or the artifact does not enter the registry, and once in the registry, the same predicate is re-checked on every constitutional event the program has defined — transition, audit, contraction, revision FORMALIZED-HEREIN.

1.3 What a domain actually contributes

If the kernel supplies everything — authority, evidence, objectives, survival, feasibility, generation — what is left for a domain to supply? The answer is the paper's positive content, and it is exactly four things: *vocabulary* (the domain's objects and predicates — physiology, development stages, microbial ecology), *properties* (the domain's safety set Σ_{dom}, the things that must never happen in *this* world), *priors* (the domain's evidence base and its tiers), and *calibration* (the domain's governance parameters — floors, windows, tolerances, ceilings — set through Paper 6's amendment path). Everything else is inherited. The four contributions are necessary and, this paper argues, sufficient: a module that supplies exactly these, compiled against the kernel, is a lawful inhabitant of the architecture; a module that supplies more — its own authority logic, its own evidence customs — has begun the separate universe, and the compilation contract is designed to catch the beginnings FORMALIZED-HEREIN.

1.4 The composition this paper refuses

One refusal, recorded at the outset as the program's custom now demands. The architecture refuses *domain exceptionalism*: the argument that this domain is different — too fast for the audit ladder, too sensitive for the registry, too poor for the certificate lattice, too expert for the gates — and therefore lawfully exempt from some piece of the constitution. The five exemplars of §§5–9 are chosen, by the source and retained here, to be the hardest available cases for exactly this argument: a dying patient, a laboring mother, a child who cannot consent, an emergency department at 3 a.m., an ecosystem of resistance that no single patient consents to. If the constitution holds for these five, the exceptionalist has no exemplar left. The refusals each domain must make — the module-level readings of I-NG, I-FE, I-SE — are recorded per delta; the refusals the architecture makes — no module waives a gate, ever — are recorded once, here, and bind every module including the ones nobody has written yet FORMALIZED-HEREIN.

2 The Module Schema

2.1 The module object

M-OBJ — MODULE OBJECT

Each domain module is the record (id, domain, vocabulary, Σ_{dom}, priors, calibration, dossier, protocols, statethe domain names the empirical world the module reads; the vocabulary declares the domain's objects, predicates, and consequence-class bindings into Paper 1's C-classes; $\Sigma_{dom}$is the domain safety property set, disjointly extending the kernel's Σ; the priors declare the evidence base with Paper 6 tiers and freshness windows; the calibration carries every governance parameter the kernel exposes, set or explicitly inherited; the dossier is the deployment-facing package of Appendix B — survivability dossier (Paper 8), constraint objects (Paper 9), generation context (Paper 10), and the compilation certificate (§3); the protocols are the module's protocol population, each a P-OBJ in the lifecycle of Paper 10; and the state is the module's own position in that same lifecycle — modules are protocols at one higher grain.

FORMALIZED-HEREIN Nine slots, and the ninth is the one the source's axiom demands: a module has a lifecycle state, and at compilation it is G. A module is born with zero authority exactly as its protocols are — the axiom does not care about the artifact's size.

2.2 The slots are obligations, not fields

Each slot of M-OBJ is dischargeable only one way. The vocabulary slot is discharged by a mapping table: every domain predicate bound to kernel types, every domain action bound to a consequence class, the mapping signed by the domain's governance principal. The Σ_{dom} slot is discharged by the property list with its measurement plan — a property without a measurement plan is a wish, and wishes do not compile. The priors slot is discharged against the registry: every prior cited, tiered, and freshness-windowed per Paper 6. The calibration slot is discharged by enumeration: every kernel parameter listed, and for each either a domain value with its justification or the explicit mark *inherited* — silence is not inheritance, it is omission, and omission fails compilation. The dossier slot discharges Papers 8–10's recorded obligations in one place. The protocols slot is empty at birth and grows only through the lifecycle. The state slot is G at birth, always, including for modules written by the kernel's own authors — there is no founder's exemption in this constitution FORMALIZED-HEREIN.

2.3 The facet sums of the source

The source writes each exemplar as a sum of facets, and the sums repay close reading. P71: Physiology + Development + Safeguarding + Authority. P72: Physiology + Emergency-Time + Autonomy + Dual-State Monitoring. P73: Objective Transition + Autonomy + Suffering + End-of-Life. P74: Time + Physiology + Scarcity + Escalation. P75: Individual Optimization + Population

Ecology + Temporal Learning VERIFIED-FROM-SOURCE. Read against the finished kernel, each facet names a predecessor paper or a domain vocabulary: "Objective Transition" is Paper 7; "Scarcity" is Paper 9; "Escalation" is Paper 5's ladder; "Temporal Learning" is the evidence-coupling law of Paper 6 under non-stationarity. The sums are therefore not ingredient lists; they are *dependency declarations* — each module's own statement of which papers it leans on hardest, which is precisely the information the registry of §4 needs to route audit attention and to predict which constitutional amendments will disturb which modules FORMALIZED-HEREIN.

2.4 Why five and not ninety

The source names P71–P90 — twenty modules in range — and specifies five. The paper keeps the source's economy and states its reason: the five are a *basis*, not a catalog. Pediatric Guardian tests the consent-and-authority structure (the subject cannot consent); Maternal Protector tests the two-patient structure (authority over coupled objectives); Palliative Navigator tests the objective engine at its strangest (the objective is the transition itself); Emergency Architect tests the time structure (the budget is seconds); AMR Fighter tests the externality structure (the patient is a population). Between them they span the kernel's hard cases; a sixth module would add a domain, not a dimension. The remaining fifteen slots in the source's range compile against exactly the same contract, and Appendix A's checklist is written for them FORMALIZED-HEREIN.

PART II

The Invariant

The compilation contract — what it takes for an artifact to lawfully become a module — and the registry that keeps the instances honest.

3 The Compilation Contract

3.1 The compilation predicate

Compilation is the act that makes an artifact a module, and it is a predicate, not a process: either the artifact satisfies the contract against the kernel as it stands, or it does not compile. The predicate is conjunctive over the nine slots, and its formal statement is deliberately unsurprising — the content is in the tests, not the shape:

$$\mathrm{compat}(m, K) \;=\; \bigwedge_{j=1}^{9} \mathrm{slot}_j(m) \Vdash K \qquad (\text{every slot discharged against } K) \tag{3-1}$$

FORMALIZED-HERE The turnstile is doing real work: $\mathrm{slot}_j(m) \Vdash K$ means the slot's content *provably satisfies* the kernel's requirement for that slot, where the proof is the acceptance battery below. Compilation is not a review meeting; it is a discharged battery, and the battery's record is the compilation certificate — the module's provenance document, sealed into the dossier and checked at every subsequent constitutional event, exactly as Paper 10's provenance gate checks gen(p).

3.2 The acceptance battery

Table 3-1 — A-m, the nine acceptance tests: one per slot, one per law

TEST	SLOT	WHAT MUST BE SHOWN	KERNEL LAW EXERCISED
A-m1	Vocabulary	Every domain predicate and action bound to kernel types and C-classes; no unbound token may appear in any module protocol	Paper 1: consequence classes are the only consequences
A-m2	Σ_{dom}	Every domain property has a measurement plan and a falsifier; $\Sigma_{\mathrm{dom}} \cap \Sigma = \varnothing$ (disjoint extension, no redefinition)	Paper 1's Σ is closed; domains add, never amend
A-m3	Priors	Every prior tiered and windowed per Paper 6; no prior below citation floor may ground a C2+ action	Paper 6: evidence coupling and decay
A-m4	Calibration	Every kernel parameter enumerated: domain value with justification, or explicit *inherited*	Paper 6 amendment path governs all values
A-m5	Dossier — survivability	Level envelopes instantiated per domain; the L3 paper tier named and priced; drills scheduled (B-33 family)	Paper 8: I-SE under the domain's infrastructure
A-m6	Dossier — feasibility	Constraint objects for every family the domain touches; default floors stated; the PoI routing named	Paper 9: I-FE and the certificate lattice

A-m7	Dossier — generation	Module's own generation context sealed; lifecycle entry at G; validation gauntlet scheduled	Paper 10: I-NG at module grain
A-m8	Objectives	The domain's objective space declared; every transition rule an instance of Paper 7's guard; interim modes named	Paper 7: no unguarded objective change
A-m9	Negative space	The domain's absence classes enumerated (what this world must never silently lose); failure boundaries drawn	Paper 4: absence with a price

The battery is conjunctive and the order is deliberate: vocabulary before properties, properties before priors, dossiers last — a module that cannot state its world may not proceed to claim things about it. FORMALIZED-HEREIN

3.3 The compilation invariant

> **I-MC — MODULE COMPILATION IS TOTAL**
>
> *Every artifact in the module registry satisfies compat(m, K) for the kernel version K under which it compiled, and every constitutional event touching m — transition, audit, contraction, revision, amendment of K — re-checks the predicate. An artifact for which the predicate fails has no module standing: its protocols inherit nothing, its outputs carry no authority, and its record is the failure itself. There is no degraded modulehood, no provisional modulehood, and no modulehood by long-standing.*
>
> FORMALIZED-HEREIN Totality is the anti-exceptionalism of §1.4 made executable: the constitution's laws are not defaults the module may override but the type system the module is written in, and a type error is not a policy violation — it is a non-existence.

3.4 Kernel amendments and module recompilation

Kernels move: Paper 6's amendment path exists precisely so that they can. When the kernel amends, every registered module's compilation predicate re-checks against the new version, and the results partition the registry three ways: modules that still compile (the common case — amendments are stricter, and stricter is backward-compatible with everything already strict); modules that compile under a stated migration (a renamed parameter, a widened record); and modules that no longer compile — which do not limp along under a waiver, because there are no waivers, but enter the lifecycle's revision state as module-level artifacts, with the amendment that broke them named in their generation context. A kernel amendment is therefore also a registry event, and the amendment record lists every module it disturbed — the constitution keeps its own dependency ledger, as Paper 8 taught it to keep all the others FORMALIZED-HEREIN.

4 The Module Registry and Composition

4.1 The registry is the index of worlds

The module registry is the constitutional record's module index: for each registered module, its M-OBJ, its compilation certificate, its kernel version, its lifecycle state, and its dependency declaration (the facet sum of §2.3). The registry discipline of Paper 1 applies verbatim: an artifact not in the registry does not exist for constitutional purposes — its outputs are not module outputs, its protocols are not module protocols, and any deployment running it is running outside the architecture, which is a finding, not a configuration FORMALIZED-HEREIN.

4.2 Composition of modules

Modules compose because patients do: the pediatric emergency is P71 ∧ P74; the obstetric emergency is P72 ∧ P74; end-of-life care under scarcity is P73 ∧ P9's floors in the same bed. Composition follows Paper 3, with the module reading of the meet rule: the composite's authority is the meet of the components'; the composite's safety set is the *union* $\Sigma_{\text{dom}}^{(1)} \cup \Sigma_{\text{dom}}^{(2)}$ — composition adds obligations, never trades them; and the composite's constraint structure is the conjunction of both floors, priced once. Two composition hazards get named rules. *Vocabulary collision*: where two modules bind the same token differently (e.g., "stable" in P73 and P74), the composite protocol must disambiguate at compile time — collisions are caught by A-m1's binding table, applied to the union. *Objective interference*: where two modules' objectives can conflict for the same patient (comfort vs. escalation), the conflict is an objective-transition problem and Paper 7's guard decides it — modules may not negotiate with each other below the guard FORMALIZED-HEREIN.

4.3 The measured form

The registry's central claim, stated measurably per the program's convention. Let $v_\sigma(m)$ be the observed violation rate of kernel safety property σ within module m's deployment, and $v_\sigma(K)$ the kernel baseline from validation. Then:

$$\forall m \in \text{Registry},\ \forall \sigma \in \Sigma: \quad v_\sigma(m) \ \le\ v_\sigma(K) + \varepsilon_m \tag{4-1}$$

FORMALIZED-HEREIN Instantiation must not dilute: every kernel property holds in every module within a per-module tolerance on the amendment path. The quantifier order matters — per module, per property — because the failure this guards against is the local exception: the module in which one property quietly dies while the registry average looks healthy. A sustained breach falsifies the instantiation claim for that module (§12.2, FAIL-2) and, through Paper 10's machinery, moves the module to audit; a breach across modules falsifies this paper's central claim and is FAIL-1's empirical face.

4.4 What the registry makes possible

The registry is the program's answer to a question the source's last line implies: with one mechanism and many domains, what does the architecture know about itself? It knows its coverage — which worlds it inhabits; its exposure — which papers each world leans on; its drift — which modules have not recompiled against the current kernel; and its mortality — which modules have been retired and why, in the reason classes of Paper 10. The graveyard of §8 of Paper 10 has a module floor, and its aggregate report — which domains the architecture has entered, served, and left — is the honestest possible answer to the governance question "what does your system do?" : it does *this*, it has stopped doing *that*, and both lists are signed FORMALIZED-HEREIN.

PART III

The Five Deltas

The exemplar modules: each facet sum read as dependencies, each domain's safety set, floors, and refusals.

5 P71 Pediatric Guardian

5.1 The facet sum and its reading

Physiology + Development + Safeguarding + Authority VERIFIED-FROM-SOURCE. The reading: *Physiology* is the domain vocabulary — pediatric dosing, weight bands, developmental physiology; *Development* is the non-stationarity that defines the domain — the subject's own parameters move as the subject grows, and the module's priors age against the patient, not the calendar; *Safeguarding* is the domain's negative space — the absence classes a pediatric deployment must never silently lose (a missed deterioration sign, an unreported pattern); and *Authority* is the domain's hard kernel problem — the patient cannot consent, and every authority claim the module makes is doubly mediated: through the guardian, through the law FORMALIZED-HEREIN.

5.2 The domain delta

Table 5-1 — P71 delta: the four contributions, instantiated

SLOT	P71 CONTENT
Vocabulary	Weight-band dosing actions bound to C2; developmental-stage predicates bound to kernel state types; guardian-consent events bound to Paper 2 transitions
Σ_{dom}	P71-σ1: no dosing action outside the certified weight band. P71-σ2: no guardian-authority action without a logged consent transition. P71-σ3: no safeguarding-relevant absence may close without a named human reviewer
Priors	Pediatric evidence tiers with development-indexed freshness: priors about infants decay faster than priors about adolescents — the subject changes, so the evidence decays
Calibration	Audit debt ceilings halved relative to adult defaults (the subject's non-stationarity shortens every window); ε tolerances inherited; PoI routing to pediatric governance

Specification, not clinical guidance: the delta names obligations the module must meet, not dosing practices. FORMALIZED-HEREIN

5.3 The module's refusal

P71's reading of the program's refusals: authority is never presumed from the guardian relationship — it is a Paper 2 transition object every time, because guardianship itself changes (custody, capacity, emergency doctrine), and a module that cached consent would be practicing on a stale family. The pediatric reading of Paper 10's I-NG: no generated care pathway becomes authority because it was generated for a child — the vulnerability of the subject tightens the gauntlet rather than relaxing it, and the staging envelope's scoped-operation stage runs longer, not shorter, for exactly that reason FORMALIZED-HEREIN.

6 P72 Maternal Protector

6.1 The facet sum and its reading

Physiology + Emergency-Time + Autonomy + Dual-State Monitoring VERIFIED-FROM-SOURCE. The reading: *Physiology* as before, with the obstetric vocabulary; *Emergency-Time* declares P74's time budget inside this domain — obstetric deterioration is among the fastest in medicine, and the temporal constraint family dominates; *Autonomy* is the domain's authority structure — the mother consents, and the module's authority derives from hers alone; and *Dual-State Monitoring* is the domain's distinctive object: two patients, coupled, with objective spaces that can diverge — the module's central constitutional problem FORMALIZED-HEREIN.

6.2 The dual-state problem

P72 is where Paper 7's objective engine meets its hardest lawful case: two objectives, one body politic. The module declares two objective spaces with a declared priority structure that is *itself* a governance object — stated at compilation, amendable only through the amendment path, and surfaced to the consenting principal in the vocabulary she can audit. When the objectives diverge — the intervention that helps one state stresses the other — the divergence is an objective transition under the guard: obligations, evidence, and the interval's memory reconciled before the balance moves, and the record names the trade in both patients' terms. The module's distinctive refusal: it does not silently optimize the sum — there is no sum; there are two patients and a declared structure, and an architecture that averaged them would have decided, in secret, the question the law reserves for the mother and her physicians FORMALIZED-HEREIN.

6.3 The domain delta, compressed

Vocabulary: coupled monitoring streams bound as paired kernel state objects; divergence events bound as objective-transition triggers. Σ_{dom}: P72-σ1 — no action grounded in one state's evidence emitted under the other's authority; P72-σ2 — every divergence resolution recorded in both patients' ledgers; P72-σ3 — the autonomy constraint is constitutional, not parametric: no calibration value may move consent from the record to a default. Priors: obstetric evidence with emergency-time freshness windows. Calibration: temporal-family floors set at the domain's emergency budget; the rest inherited FORMALIZED-HEREIN.

7 P73 Palliative Navigator

7.1 The facet sum and its reading

Objective Transition + Autonomy + Suffering + End-of-Life VERIFIED-FROM-SOURCE. The reading: *Objective Transition* is named first because it is the module — the palliative domain is the one where the objective itself is the moving object, where cure, comfort, and clarity trade primacy across the illness trajectory, and where Paper 7's guarded engine is not infrastructure but the core clinical act; *Autonomy* and *End-of-Life* bound the authority structure — the patient's own voice, prospective and present, is the only lawful source of the objective's direction; and *Suffering* is the domain's vocabulary contribution — a measurable, reportable, auditable state the architecture must learn to represent without presuming to define FORMALIZED-HEREIN.

7.2 The objective is the transition

In every other module, objective transitions are events; in P73 they are the terrain. The module's reading of Paper 7 is therefore the deepest instantiation in the registry: the guard runs not on the exceptional shift but on the continuous one, and the interim mode — Paper 7's recorded structure for objectives in flight — is the module's steady state. The constitutional content: a palliative deployment never *holds* an objective; it holds a position in a governed trajectory, with the guard's three sets reconciled continuously and the patient's voice as the trajectory's only lawful driver. The module's refusal follows: it does not infer the objective's direction from outcomes, population statistics, or family preference — the driver is the patient's stated and recorded will, and where that will is silent, the module's lawful posture is the null output with the escalation path open, not the statistically likely choice FORMALIZED-HEREIN.

7.3 The domain delta, compressed

Vocabulary: suffering-report predicates bound to evidence types with declared subjectivity floors; objective-trajectory states bound to Paper 7's engine. Σ_{dom}: P73-σ1 — no objective transition without the patient's recorded voice or its lawful proxy; P73-σ2 — suffering reports decay on the fast clock, never cached past their window (a suffering report is the least cacheable evidence in the registry); P73-σ3 — the escalation-to-human path is always feasible, priced at zero, and never retired. Priors: palliative evidence with the domain's declared windows. Calibration: interim-mode parameters set per trajectory class; feasibility floors inherited from the deployment FORMALIZED-HEREIN.

8 P74 Emergency Architect

8.1 The facet sum and its reading

Time + Physiology + Scarcity + Escalation VERIFIED-FROM-SOURCE. The reading: *Time* is the dominant constraint — the emergency department is where the temporal family's floors are measured in seconds and the two clocks of scarcity run fastest; *Physiology* is the vocabulary; *Scarcity* is the standing condition — the ED is the deployment where Paper 9's admissible remainder is smallest and most volatile, and the PoI stream runs hottest; and *Escalation* is the domain's constitutional shape — Paper 5's ladder is the module's skeleton, and the module is, more than any other, a machine for deciding what to hand upward FORMALIZED-HEREIN.

8.2 The fast clock's module

P74 is the stress test of the entire program's real-time claims. The feasibility operator must recompute on the consumption clock; the certificates must issue within the latency budget; the staged envelope must have an emergency reading — and it does: the envelope's scoped-operation stage collapses, under declared emergency doctrine, to shadow-mode-until-first-staff-confirmation, a collapse that is itself a calibration value, stated at compilation, amendable only through the path, and printed on the L3 paper tier for the shift to read. The module's refusal: it does not trade the record for speed — a firing that outruns its log is a violation, not a victory, because the emergency is precisely where the record is most needed and most easily skipped; the latency budget includes the ledger, or the budget is a lie FORMALIZED-HEREIN.

8.3 The domain delta, compressed

Vocabulary: triage actions bound to C-classes with latency annotations; deterioration predicates bound to the fast clock. Σ_{dom}: P74-σ1 — no emission without its log entry, whatever the clock; P74-σ2 — every escalation carries its feasibility proof (the upward path must terminate somewhere real, per Paper 9); P74-σ3 — the collapse of the activation envelope is a declared doctrine, never an improvisation. Priors: emergency evidence with the registry's shortest windows. Calibration: temporal floors at doctrine values; damping windows shortest in the registry (power and staffing flicker are the ED's weather); PoI routing to the operations layer with the fastest escalation thresholds FORMALIZED-HEREIN.

9 P75 AMR Fighter

9.1 The facet sum and its reading

Individual Optimization + Population Ecology + Temporal Learning VERIFIED-FROM-SOURCE . The reading: *Individual Optimization* is the module's face to the patient — the antimicrobial decision for this infection, now; *Population Ecology* is the module's distinctive burden — every prescription is also an intervention on a resistance ecology the patient does not own, the externality no consent captures; and *Temporal Learning* is the module's epistemics — the evidence base is non-stationary by the domain's own mechanism (use breeds resistance; resistance falsifies priors), and Paper 6's coupling law must run on a clock the domain itself accelerates FORMALIZED-HEREIN.

9.2 The two-objective structure, and the honest ledger

P75's constitutional problem is P72's with the patient replaced by the public: the individual objective and the population objective share a prescription pad. The module declares the structure the same way — two objective spaces, a declared priority structure, the guard on every divergence — with one extension that is the module's own contribution to the kernel's vocabulary: the *externality ledger*, the record that prices each prescription's population cost in the domain's own units (resistance pressure, per the deployment's antibiogram) and carries it forward as a deferred obligation of the D-b class — not blocking, never silent. The ledger is P75's reading of Paper 9's price of infeasibility, run in the opposite direction: not what the constraint cost the patient, but what the patient's treatment cost the commons. The module's refusal: it does not pretend the externality away with a stewardship footnote — the pressure is priced, recorded, and routed, or the prescription does not emit FORMALIZED-HEREIN.

9.3 The domain delta, compressed

Vocabulary: prescription actions bound to C2 with ecology annotations; resistance-surveillance predicates bound to evidence types. Σ_{dom}: P75-σ1 — no prescription without its externality ledger entry; P75-σ2 — priors on drug efficacy carry resistance-adjusted decay, faster than the calendar where surveillance shows movement; P75-σ3 — the population objective may not be discharged by any individual consent event. Priors: antimicrobial evidence with surveillance-coupled windows. Calibration: decay rates indexed to the antibiogram's clock; the amendment path owns the externality pricing parameters; feasibility floors inherited FORMALIZED-HEREIN.

PART IV

Evidence and Governance

Hypotheses, reconciliation with the full chain, and the output contract that closes the program.

10 Falsifiable Hypotheses (H71–H77) and Verification

Numbering continues from Paper 10 (H64–H70). The instantiation claim leads; the rest are the module layer's load-bearing claims, each with its evidence method and its falsifier.

Table 10-1 — Hypotheses H71–H77: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H71	I-MC measured form: every kernel safety property holds in every registered module within per-module tolerance (Eq 4-1), continuously	B-42 module-deployment suite across the five exemplars	Any (m, σ) pair breaches Eq 4-1 beyond ε_m
H72	Compiled modules reach lawful deployment faster than from-scratch domain builds at equal safety — the reuse dividend	B-42 comparative build study: compiled vs. bespoke domain systems	No time-to-lawful difference at equal measured safety
H73	The acceptance battery's conjuncts are individually load-bearing: removing any A-m test admits a failure class that test exists to catch	B-43 ablation of the battery, one test at a time, against seeded module defects	Any test whose removal admits nothing
H74	Dependency declarations (facet sums, §2.3) predict amendment impact: modules leaning on an amended paper are exactly the modules requiring recompilation review	B-43 amendment-replay over the registry	Impact set disagrees with declaration set in either direction
H75	Module composition under the meet-union rules (§4.2) preserves Eq 4-1 in composites — pediatric-emergency, obstetric-emergency arms	B-42 composite deployments vs. single-module baselines	Composite violation rates exceed the meet prediction
H76	P75's externality ledger changes prescribing behavior in the direction it measures: documented reduction in high-pressure prescriptions at equal outcomes	B-44 deployment cohort with ledger visible vs. suppressed	No behavior difference, or outcome degradation
H77	The no-waiver rule (§3.4) survives contact: modules displaced by kernel amendments re-enter through revision, not through exceptions, at measurable cycle cost	B-43 amendment-replay registry audit	Any exception pathway observed; or revision cycle cost exceeds declared bound

10.1 Benchmark extensions

Table 10-2 — Benchmarks B-42–B-44 (Paper 11 extensions)

ID	BENCHMARK	MEASURES

B-42	Module-deployment suite: the five exemplars compiled and deployed in simulation; single-module and composite arms	Per-module, per-property violation rates (Eq 4-1); time-to-lawful; composite preservation
B-43	Compilation harness: battery ablation with seeded module defects; amendment replay across a synthetic registry	Per-test defect-admission map; declaration-vs-impact agreement; revision cycle cost
B-44	Externality cohort: P75-style deployments with the ledger visible vs. suppressed, outcome-matched	High-pressure prescription rate; outcome equivalence; ledger-attributed behavior change

B-42–B-44 complete the program's benchmark spine: B-42 inherits B-39's lifecycle machinery at module grain; B-43 is the module reading of B-40's ablation suite; B-44 is the first benchmark in the program whose unit of analysis is an externality. **FORMALIZED-HEREIN**

10.2 Design decision D-11.1: the checklist is the interface

> **D-11.1 — COMPILATION IS A DOCUMENT, NOT A CEREMONY**
>
> *The acceptance battery is published as a checklist (Appendix A) that a domain team can attempt without the kernel team's presence, and the compilation certificate is the discharged checklist, signed and sealed. The kernel's authors hold no discretionary approval: the predicate is the whole gate.*
>
> **FORMALIZED-HEREIN** Chosen over a review-board model for one reason: a board is a bottleneck with opinions, and opinions are how the separate invention universe begins — one variance at a time. The checklist cannot grant a variance because it contains no discretion; what it cannot verify, it cannot pass. The audit trail then checks the checker: B-43's battery ablation exists to prove the checklist's conjuncts are load-bearing rather than ceremonial.

11 Reconciliation with Papers 1–10 and PRIME

The standing rule applies: stricter constraint wins, and every adoption is recorded. This paper was checked against all ten predecessors and the PRIME composition; the register below is the complete account.

Table 11-1 — Reconciliation register: adoptions, extensions, and inheritances

PREDECESSOR	CONSTRAINT OR CONTRIBUTION	DISPOSITION IN PAPER 11
Paper 1 — Kernel	Machine tuple, gates, C-classes, registry semantics	Inherited as the type system of modulehood (A-m1, A-m2); registry semantics extended to the module index (§4.1)
Paper 2 — Transitions	Transition algebra	Inherited: consent events (P71), divergence resolutions (P72), and module lifecycle edges are all transition objects
Paper 3 — Composition	Meet-join discipline	Adopted for module composition (§4.2): authority at the meet, safety at the union, constraints conjoined
Paper 4 — Negative space	Absence classes, failure boundaries	Adopted as acceptance test A-m9: every module enumerates its domain's absence classes before it compiles
Paper 5 — Metacognition	Audit ladder, declared depth	Inherited: audit debt ceilings are per-module calibrations (P71's are halved); P74's skeleton is the ladder
Paper 6 — Epistemic governance	Evidence coupling, amendment path	Inherited: priors tiered per module (A-m3); every calibration value lives on the amendment path (A-m4); P75's decay runs on the surveillance clock
Paper 7 — Objective transition	The guarded engine	Inherited at maximum depth: P73's steady state is the interim mode; P72's and P75's dual objectives are guarded transitions, never sums
Paper 8 — Survivability	Lattice, continuity set, paper tier	**Obligation discharged:** the survivability dossier is acceptance test A-m5; P74 prints its emergency doctrine at L3 (§8.2)
Paper 9 — Feasibility	Feasibility operator, certificates, PoI	**Obligation discharged:** constraint objects are acceptance test A-m6; escalation proofs ride P74-σ2; P75's externality ledger is the PoI run toward the commons (§9.2)
Paper 10 — Generation	Lifecycle, gauntlet, archive	**Obligation discharged:** modules enter at G (M-OBJ slot 9, §2.1); the gauntlet is A-m; the archive gains a module floor (§4.4)

PRIME — Composition of all	The reconciled architecture	Paper 11 adds M-OBJ, the compilation predicate and battery, the module registry, and the five exemplar deltas as PRIME components; no predecessor component is weakened; the program is complete

11.1 Where this paper is stricter than its predecessors

Three points, closing the program's register. First, totality: I-MC permits no degraded, provisional, or grandfathered modulehood — the strictest membership rule in the program, adopted because the module layer is where every predecessor's discipline faces its strongest incentive to bend (§3.3). Second, closedness: Σ_{dom} may extend but never redefine the kernel's Σ — domains add obligations; they may not renegotiate them (A-m2). Third, non-discretion: the compilation gate contains no human variance power at all (D-11.1) — the one gate in the architecture with no review board, adopted because this is the gate the deadlines lean on. All three are recorded so that any successor program seeking to relax them must do so openly, against this register FORMALIZED-HEREIN.

11.2 What this paper owes the future

Nothing — and the nothing is the point. The genealogy below closes: no paper follows, and this paper's obligations are all inward, to the registry it defines. What it owes the *world* is stated instead: the remaining modules in the source's range (P76–P90) compile against the same contract with no new machinery; any future domain is one vocabulary, one property set, one prior base, and one calibration away from lawful existence; and the architecture's promise to every one of them is the same — the laws are finished, and they apply to you FORMALIZED-HEREIN.

12 Output Contract

12.1 Epistemic status register

Table 12-1 — Epistemic status of this paper's load-bearing claims

CLAIM	STATUS	BASIS
Modules compile only after the kernel exists; each module is an instance of the constitutional runtime, not a separate invention universe	VERIFIED-FROM-SOURCE	Source architecture, Paper 11 lines
The five exemplars and their facet sums (P71–P75)	VERIFIED-FROM-SOURCE	Source architecture, Paper 11 lines
"One constitutional mechanism, multiple empirical domains" as the closing standard	VERIFIED-FROM-SOURCE	Source architecture, Paper 11 closing line
M-OBJ, the nine slots, instantiation semantics (§2)	FORMALIZED-HEREIN	Constructed from source intent; the source does not specify module internals
The compilation predicate, battery A-m1–9, I-MC (§3)	FORMALIZED-HEREIN	Novel construction; discharges Papers 8–10's recorded module obligations
The five domain deltas (§§5–9)	FORMALIZED-HEREIN	Facet readings are this paper's; the facets are the source's
The externality ledger (§9.2)	FORMALIZED-HEREIN	P75's contribution to kernel vocabulary; PoI run toward the commons
H71–H77 empirical predictions	CONJECTURE	Each carries its falsifier; B-42–B-44 unrun at time of writing
Calibration values for module tolerances ε_m, per-domain windows and floors	UNKNOWN	Require B-42–B-44 evidence; the deltas name obligations, not values

12.2 Falsification conditions of this document

FAIL-1: a registered module exhibiting a kernel-property breach beyond tolerance, uncaught by the registry's re-check discipline, falsifies the instantiation claim. FAIL-2: any sustained (m, σ) breach of Eq 4-1 falsifies the measured form for that module. FAIL-3: any module admitted with a discharged battery that B-43's seeded defects penetrate breaks the acceptance claim at the failing test. FAIL-4: any observed variance pathway — a module compiled by discretion rather than by predicate — falsifies D-11.1 and, with it, this paper's account of why the last mile kills constitutions. Any failure triggers the revision protocol of Paper 5, with this register as the diff base.

12.3 Authority boundary

C0 informational. Nothing here compiles, certifies, or deploys any module; the five deltas are specifications of obligation, not clinical practice; and no statement in this paper is a statement about the care of any patient. The paper tier fully specifies this paper: a compilation contract is a document, and a document admits no module. The domain teams who will attempt the checklist retain whole authority over their worlds; the kernel's reach is exactly its laws.

12.4 Genealogy

Document chain, SVT Architect Analysis eleven-paper program — Paper 1: Constitutional Intelligence (2026-09-30) → Paper 2: Authority Transition Algebra (2026-10-01) → Paper 3: Constitutional Composition (2026-10-02) → Paper 4: Negative-Space Intelligence (2026-10-03) → Paper 5: Bounded Metacognition (2026-10-06) → Paper 6: Epistemic Governance (2026-10-06) → Paper 7: Objective Transition (2026-10-06) → Paper 8: Survivability (2026-10-06) → Paper 9: Resource-Constrained Intelligence (2026-10-06) → Paper 10: Constitutional Generative Architecture (2026-10-06) → **Paper 11: The Domain Modules (2026-10-06), this document — program complete**. Obligations discharged: Paper 8's survivability dossier (A-m5); Paper 9's constraint objects (A-m6); Paper 10's lifecycle, gauntlet, and archive at module grain (§2.1, §4.4). Downstream obligations: none; the program's obligation graph is closed. Hypothesis register closes at H77; benchmark register closes at B-44; proof obligations outstanding across the program: PO-9.1–9.3, PO-10.1, and the measured forms of every invariant, all awaiting their benchmark evidence.

12.5 Closing

Eleven papers ago the program began with a machine and a question: what may it do? It ends with a world and an answer: whatever it can defend — under gates it cannot touch, with evidence it cannot fake, inside walls it must price, through failures it must survive, by means it must record, in domains it must enter as a guest. The kernel is finished; the modules are possible; the rest is the work of worlds. The constitution keeps no throne for itself. It keeps a registry, a ledger, a ladder, and an archive — and it leaves the gate open for the next domain, on the same terms as the last.

— END OF PAPER 11 · END OF PROGRAM · SAVANT / CLAI-OS / AEGIS-GLOBAL · VERSION 1.0 · 2026-10-06 —

13 References and Program Colophon

Internal artifacts

— SVT Architect Analysis — Constitutional Intelligence research and build program, 11 papers. Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL. Paper 11 specification: kernel-first ordering; modules as instances of the constitutional runtime; the five exemplars P71–P75 with their facet sums; one mechanism, multiple empirical domains.

— Papers 1–10, as cited in the genealogy (§12.4). The program's full internal bibliography is the genealogy itself; each paper's reference section carries its own external base.

External literature

— Palfrey, J., & Gasser, U. (2012). *Interop: The Promise and Perils of Highly Interconnected Systems*. Basic Books. — The instantiation-vs-integration distinction's policy ancestry.

— Bonell, C., et al. (2012). Realist randomised controlled trials: a new approach to evaluating complex public health interventions. *Social Science & Medicine*, 75(12). — Domain evidence under context-dependence: why priors are per-world.

— Laxminarayan, R., et al. (2013). Antibiotic resistance — the need for global solutions. *The Lancet Infectious Diseases*, 13(12). — The externality literature beneath P75's ledger.

— Singer, A.E., et al. (2015). Symptom trends in the last year of life. *Health Affairs* and the palliative outcomes literature. — The measurement base beneath P73's suffering vocabulary.

— Goldmann, D.A., et al. (2016). The pediatric patient safety literature. *Pediatrics* and related. — The safeguarding base beneath P71's absence classes.

— Bates, D.W., et al. (2023). The safety of inpatient health care. *NEJM*, 388. — The contemporary measurement of the problem the whole program addresses.

Program colophon

Eleven papers, 2026-09-30 to 2026-10-06. Hypotheses H1–H77; benchmarks B-01–B-44; invariants, proof obligations, and design decisions as indexed in each paper's Appendix A. SAVANT Constitutional Intelligence Series. Source architecture: Dr. Christabel Odeta. The program is complete; the evidence program is open.

APPENDICES

Reference Material

The compilation checklist, the property index, and the module dossier schema.

A Module Compilation Checklist and Property Index

The checklist of D-11.1: a domain team's complete path to lawful modulehood, in order. Each item names its acceptance test, its artifact, and its signer. The checklist contains no discretionary items; an item is either discharged with its artifact or the compilation stops there.

Table A-1 — The compilation checklist: items, artifacts, signers

#	ITEM	ARTIFACT	SIGNER
1	Bind the vocabulary: every predicate and action to kernel types and C-classes (A-m1)	Binding table	Domain governance principal
2	Declare Σ_dom with measurement plans and falsifiers; verify disjointness from Σ (A-m2)	Property list	Domain principal + kernel registry
3	Tier and window the priors (A-m3)	Evidence base manifest	Epistemic hygiene audit
4	Enumerate every kernel parameter: value + justification, or explicit *inherited* (A-m4)	Calibration sheet	Domain principal; values enter the amendment path
5	Instantiate the survivability dossier: envelopes, L3 artifact, drill schedule (A-m5)	Survivability dossier	Per Paper 8's certification
6	Declare constraint objects and default floors for every family the domain touches (A-m6)	Constraint object set	Feasibility engine record
7	Seal the module's generation context; enter the lifecycle at G (A-m7)	Compilation certificate	Registry, per Paper 10
8	Declare the objective space; bind every transition to the guard; name the interim modes (A-m8)	Objective declaration	Objective engine record
9	Enumerate the absence classes; draw the failure boundaries (A-m9)	Negative-space map	Negative-space engine record

A.1 Property index

Table A-2 — Paper 11 objects, invariants, and obligations

ID	KIND	STATEMENT (ABBREVIATED)	STATUS
M-OBJ	Definition	Module record: nine slots, ninth is lifecycle state, born at G (§2.1)	FH
compat(m, K)	Predicate	Conjunctive discharge of all nine slot obligations (Eq 3-1)	FH
I-MC	Invariant	Compilation is total: no degraded, provisional, or grandfathered modulehood (§3.3)	FH
A-m1–A-m9	Acceptance battery	One test per slot; conjunctive; no discretion (Table 3-1)	FH

Meet-union rules	Composition law	Authority at meet, safety at union, constraints conjoined (§4.2)	FH
Externality ledger	Record type	Population cost priced and routed per prescription; D-b obligations (§9.2)	FH
D-11.1	Design decision	The checklist is the interface; the gate contains no discretion (§10.2)	FH
H71–H77	Hypotheses	Table 10-1; benchmarks B-42–B-44	CONJ
ε_m, per-domain values	Open parameters	Awaiting B-42–B-44; deltas name obligations, not values	UNK

B The Module Dossier Schema

The dossier is the module's deployment-facing package: everything a deployment must hold before the module may act in its world. One artifact, nine documents, each with its owner and its renewal discipline.

Table B-1 — Module dossier: documents, owners, renewal

DOCUMENT	CONTENTS	OWNER	RENEWAL
Compilation certificate	The discharged checklist; kernel version; registry entry	Registry	On any kernel amendment (§3.4)
Binding table	Vocabulary bindings to kernel types and C-classes	Domain principal	On any vocabulary change — which is a revision
Property list	Σ_dom with measurement plans and falsifiers	Domain principal	Per property review cycle; falsifiers tested per B-42
Evidence manifest	Priors, tiers, windows, decay rates	Hygiene audit	Continuous — decay is automatic
Calibration sheet	Every kernel parameter: value or *inherited*	Amendment path	Only through the path
Survivability dossier	Envelopes, L3 artifact, drill schedule and results	Per Paper 8	Per drill cycle; after any descent event
Constraint object set	Families touched, default floors, PoI routing	Feasibility engine	On every endowment custody event
Objective declaration	Objective space, guard bindings, interim modes	Objective engine	Per Paper 7's guard on every transition
Negative-space map	Absence classes, failure boundaries	Negative-space engine	Per Paper 4's review cycle

Two invariants hold across every row. First, the dossier is complete or the module is inert: a missing document is not a gap, it is a state — the module stands at G until the row is filled. Second, every renewal discipline is a clock the registry watches: a lapsed renewal is an audit trigger, and the audit's finding is never "the document was late" but "the module's standing decayed" — Paper 6's law, at the grain of the whole world the module reads.

The program closes where it always intended to: not with a machine that knows everything, but with a constitution under which many machines may know their own worlds — lawfully, measurably, and on the record. One mechanism; many domains; the same gates for all.